William Kwako

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EDUCATION

M.S., Computer Science, Georgia Tech, Atlanta, GA — Expected graduation date: August 2026
B.A., Physics, Willamette University, Salem, OR — May 2019

PROFESSIONAL EXPERIENCE

Technical Solutions Engineer – Epic Systems Corporation — December 2019 – December 2022

- Supported clients with deployment and maintenance of healthcare software in the Claims and Remittance department
- Headed the Debugging Task Force; responsibilities included teaching new hires how to use internal debugging tools, revising the curriculum, and delegating assignments to task force members
- Conducted quality assurance on multiple development projects using PQA framework
- Created searches with M to retrieve information from Healthcare IT databases for issue resolution and data analysis

Teacher's Assistant – Willamette University, Physics Department — January 2019 – May 2019

- Guided Students through computational and experimental Introductory Physics 2 laboratory exercises
- Answered questions about class material and evaluated students' performance

Intern – Willamette University, Physics Department — May 2018 – July 2018

- Analyzed frequency and location of baryon acoustic oscillations in 2MRS
- Located two BAOs with signals and characteristics agreeing with results of previous studies
- Worked closely with professor and peers to build computer for parallel GPU processing

Volunteer – UCSB Department of Cosmology — July 2017

- Assisted research staff in measuring the maximum intensity of light receivable by photodiodes
- End goal of sending small satellites to Proxima Centauri propelled by lasers and a solar sail system

PROJECTS

- Improving Students' Code Comprehension Ability Through LLM-Generated Practice Problems
  Educational technology project
  - Built a webapp that leverages LLMs to generate programming exercises, aiming to improve students' code comprehension skills
  - **Tools**: Python, Django, JavaScript, HTML/CSS, OpenAI and Anthropic APIs
- Ravens' Progressive Matrices Solver
  Knowledge-based AI project
  - Created a Python program that applies computer vision and pattern analysis to automatically solve visual reasoning puzzles
  - **Tools:** Python
- Looking for the BAO Signal in the 2MRS Using the Wavelet Transformation
  Senior Thesis, Willamette University

- Applied wavelet transformations to identify baryon acoustic oscillations (BAOs) in galaxy distribution data from the 2MASS Redshift Survey (2MRS)
- **Tools**: Python

- Squire of the Crescent Realm
  Video game development project
  - Collaborated in a development team to design and implement a 3D adventure game using Unity and C#
  - **Tools**: C#, Unity, Git

POSTERS AND PRESENTATIONS

- *Improving Students' Code Comprehension Ability Through LLM-Generated Practice Problems*
  Poster Presentation – National Council on Measurement in Education (April 2026)
- *Looking for the BAO Signal in the 2MRS Using the Wavelet Transformation*
  Oral Presentation: Willamette-Linfield intercollegiate symposium (May 2019)
- *Looking for the BAO Signal in the 2MRS Using the Wavelet Transformation*
  Poster Presentation: 223rd Meeting of the American Astronomical Society (January 2019)
- *Looking for the BAO Signal in the 2MRS Using the Wavelet Transformation*
- Poster Presentation: Murdock College Science Research Conference (November 2018)

TECHNICAL SKILLS

Programming

- Python
- Java
- M
- SQL
- C#
- CSS
- HTML

Software

- Microsoft 365 Suite
- LaTeX
- Git
- Unity
- **A**ndroid Studio

Hardware

- Building computers
- Arduino-based electronics

HONORS AND AWARDS

Jason Lee Scholarship — August 2015 – May 2019
Willamette University College Honors — February 2016 – May 2019
Murdock Charitable Trust College Research Program for the Natural Sciences ($5500) — June 2018 – August 2018